Chapter 7, Problem 38P

Problem

A circuit is described by

$$\frac{di}{dt} + 3i = 2u(t)$$

Find $i(t)$ for $t > 0$ given that $i(0) = 0$.

Step-by-step solution

Step 1 of 3

The differential equation of a circuit is,

$$\frac{di}{dt} + 3i = 2u(t)$$

$$\frac{di}{dt} = 2u(t) - 3i$$

$$\frac{di}{dt} = -(3i - 2)$$

$$\frac{di}{3i-2} = -dt$$

Step 2 of 3

Apply integration on both sides.

$$\int_{i(0)}^{i(t)} \frac{di}{3i-2} = -\int_0^t dt$$

$$\left[\frac{\ln(3i-2)}{3}\right]_{i(0)}^{i(t)} = -[t]_0^t$$

$$\frac{1}{3}\ln\left(\frac{3i(t)-2}{3i(0)-2}\right) = -t \quad \ldots\ldots (1)$$

Step 3 of 3

Substitute 0 for $i(0)$ in equation (1).

$$\ln\left(\frac{3i(t)-2}{3(0)-2}\right)=-3t$$

$$\ln\left(\frac{3i(t)-2}{-2}\right)=-3t$$

$$\frac{3i(t)-2}{-2}=e^{-3t}$$

$$3i(t)-2=-2e^{-3t}$$

$$3i(t)=2-2e^{-3t}$$

$$i(t)=\frac{2}{3}-\frac{2}{3}e^{-3t}$$

$$i(t)=\frac{2}{3}\left(1-e^{-3t}\right)\ \mathrm{A}$$

Therefore, the expression for $i(t)$ is $\boxed{\frac{2}{3}\left(1-e^{-3t}\right)\ \mathrm{A}}$.